input voltage is applied to the circuit and constant output voltage V_o is taken across the load resistance R_L which is connected in parallel with Zener diode. The output voltage is maintained constant as long as the input voltage is greater than V_Z.

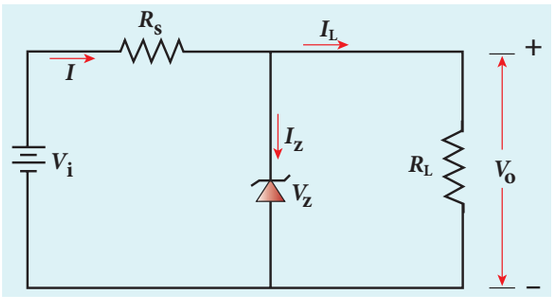


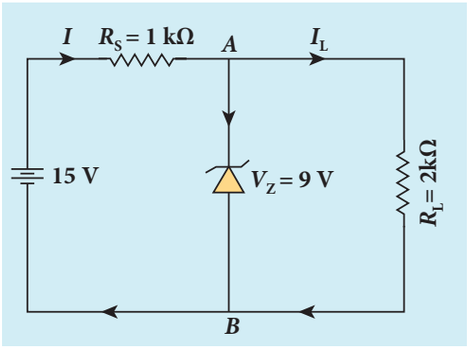
Figure 10.21 Circuit to study voltage regulation by Zener diode

If the input DC voltage is increased, the Zener current increases thereby increasing current through R_s and the voltage drop across R_s is also increased. The increased current flows through the diode without affecting the I_L. Since Zener diode is operated in the breakdown region, the Zener breakdown voltage across the diode is nearly constant even though the reverse bias current through the diode increases considerably. The increase in input voltage is dropped across R_s and hence it is also called dropping resistance. Because of the parallel connection, the voltage across R_L is also equal to Zener breakdown voltage which is taken as constant output voltage V_0.

If the input DC voltage is decreased, the diode takes a smaller current and the voltage drop across R_s is reduced. Thus, the output voltage V_0 remains constant. To sum up, if there is any change in input voltage, the voltage drop across R_s changes accordingly. But the voltage across Zener diode or voltage across R_L remains constant. Thus the Zener diode acts as a voltage regulator.

EXAMPLE 10.3

Find the current through the Zener diode when the load resistance is 2 kΩ. Use diode approximation.



Solution

Voltage across AB, $V_Z = 9V$

Voltage drop across $R_S = 15 - 9 = 6V$

Therefore current through the resistor R_S,

$$I = \frac{6}{1\times10^3} = 6 \text{ mA}$$

Voltage across the load resistor, $V_{AB} = 9V$

Current through load resistor,

$$I_L = \frac{V_{AB}}{R_L} = \frac{9}{2\times10^3} = 4.5 \text{ mA}$$

The current through the Zener diode,

$$I_Z = I - I_L = 6\text{mA} - 4.5\text{mA} = 1.5 \text{ mA}$$

10.3.7 Optoelectronic devices

Optoelectronics deals with devices which convert electrical energy into light and light into electrical energy using semiconductors. Optoelectronic device is an electronic device which utilizes light for useful applications. We will discuss some important optoelectronic devices namely, light emitting diodes, photo diodes and solar cells.

i) Light Emitting Diode (LED)

LED is a *p-n* junction diode which emits visible or invisible light when it is

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